

ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of Program: B Tech (Biotechnology)

Semester: Seven

Name of Paper: Python

Paper Code: TBT-704

Time: 3 Hrs]

[Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

1.	20 Marks	CO-3
(a)	Write a Python program that uses both single-line and multi-line comments to describe the steps in solving a simple problem. What is the importance of comments in Python?	
(b)	Write a Python program that accepts two numbers from the user, performs the following arithmetic operations on them: addition, subtraction, multiplication, and division, and then prints the results.	
(c)	Create a Python program to demonstrate the use of different data types. Assign values to variables of types: integer, float, string, and boolean, and print them along with their type using the <code>type()</code> function.	
2.	20 Marks	CO-5
(a)	Evaluate the use of <code>while</code> loops for a program that computes the factorial of a number. What are the risks of using <code>while</code> loops in this case, and how can you avoid infinite loops? Suggest an alternative approach using a <code>for</code> loop and justify your recommendation.	
(b)	Analyze a function that computes the Fibonacci series and prints the result up to the n th term. Critically evaluate whether this function can handle large values of n efficiently. Suggest improvements, such as using memoization, to optimize performance.	
(c)	Evaluate a Python implementation of the Fibonacci sequence that uses recursion. Discuss the potential inefficiencies of the recursive approach and suggest improvements, such as using an iterative method or memoization, to handle larger values of n .	
3.	20 Marks	CO-4
(a)	Given a list of integers, write a Python function that performs the following operations: 1. Replaces the first occurrence of a specified element with a new value. 2. Inserts a new element at a specified index.	

	3. Removes an element from the list by its value. Analyze the time complexity of each operation and discuss which operation is most efficient for large lists.	
(b)	You are tasked with writing a program to remove duplicates from a list. Analyze the efficiency of using a set to remove duplicates compared to using a loop with conditional checks. Which approach is better for large lists and why?	
(c)	Given the dictionary <code>my_dict = {"a": 1, "b": 2, "c": 3}</code> , write a Python program that performs the following tasks: 1. Adds a new key-value pair. 2. Replaces the value of an existing key. 3. Removes a key-value pair. 4. Traverses the dictionary to print all keys and values. Analyze the time complexity of these operations and discuss the advantages of using dictionaries over other data structures such as lists and tuples.	
4.	20 Marks	
(a)	Explain the purpose of a Python function and describe the difference between a function definition and a function call. Provide an example of each.	
(b)	What are function arguments in Python? Describe the difference between positional arguments and keyword arguments, and provide an example of a function that uses both types of arguments.	CO-2
(c)	What is the purpose of importing a module in Python? Explain how you would import a module and use a function from that module in your program.	
5.	20 Marks	
(a)	What is Biopython, and what is its primary purpose in biology or bioinformatics?	
(b)	What file formats are commonly used in Biopython to store sequence data, and which module in Biopython is used to parse these formats?	CO-1
(c)	How can you compare two sequence objects in Biopython to check if they are identical? Explain with example.	